Fiscal Year:	FY 2011	**Task Last Updated:**	FY 09/08/2011
PI Name:	Brainard, George C. Ph.D.		
Project Title:	Optimizing Light for Long Duration Space Exploration		
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Program/Discipline:	NSBRI		
Program/Discipline--Element/Subdiscipline:	NSBRI--Human Factors and Performance Team		
Joint Agency Name:	**TechPort:**		Yes
Human Research Program Elements:	(1) **BHP**:Behavioral Health & Performance (archival in 2017)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	19107-5083	**Congressional District:**	1
Comments:			
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No. of Bachelor's Candidates:	1	**Monitoring Center:**	NSBRI
Contact Monitor:		**Contact Phone:**	
Contact Email:			
Flight Program:			
Flight Assignment:	NOTE: End date changed to 12/31/2012 per NSBRI (Ed., 8/29/2012) NOTE: End date changed to 9/30/2012 per NSBRI (Ed., 1/27/2012)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Lockley, Steven (Brigham and Women's Hospital) Hanifin, John (Thomas Jefferson University) Pineda, Maria (Thomas Jefferson University)		
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Task Description:

The goal of this project is to determine the optimal lighting for use as a countermeasure to sleep and circadian disruption for astronauts. Sleep deficits that diminish alertness, cognitive ability, and psychomotor performance are a serious risk factor during space missions. Currently, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). Despite their use, more than half of the astronauts still get only between 3 and 6 hours of sleep per night during spaceflight (Barger et al., 2009).
Light is the primary stimulus for human circadian regulation. On Earth, humans have a 24 h day/night cycle to maintain healthy circadian entrainment. In space, astronauts must contend with either rapidly changing or severely disrupted day/night cycles and must work in a dimly lit spacecraft interior with few windows. Bright white light has been implemented as a pre-launch countermeasure but has yet to be used during spaceflight. Providing sufficient light intensities to work areas in the ISS as well as future vehicles and habitats raises several concerns, among which are heat production, energy consumption, and up-mass. Improving these factors requires a better understanding of how different light sources regulate the human circadian system. From these data it may be possible to optimize astronaut and ground crew light exposure both as a stimulus for vision as well as a countermeasure for sleep and circadian disruption during space missions.

Currently, NASA uses white fluorescent light for interior illumination of the ISS at relatively low intensities, and for a pre-launch lighting countermeasure for circadian disruption at much higher intensities. In the previous funding period, our NSBRI research was supported to determine if those fluorescent lamps have increased efficacy for melatonin suppression and circadian phase-shifting when they are enriched in the blue portion of the spectrum. Philips Lighting, an NSBRI industry partner, provided prototype blue-enriched lamps and exposure systems for that study. In addition, we began characterizing the circadian potency of ambient extraterrestrial light that astronauts will encounter on the surfaces of the Moon and Mars.

To develop an in-flight lighting countermeasure for space exploration, it is vital to determine the sensitivity of astronauts' circadian systems to both ambient and artificial lighting stimuli. In the current period of research, we are shifting our efforts from testing fluorescent light sources to testing solid-state lights. Artificial illumination for future space vehicles and habitats will be provided predominantly by light emitting diodes (LEDs). The following list identifies the progress on our specific aims.

1. Determine the neuroendocrine, alerting, and circadian potency of light emitted by the white LEDs being considered for re-lamping the ISS as well as lighting future vehicles and habitats. For this aim, we have had four 4' x 4' white LED lighting systems installed in our laboratory. These systems emit white light with CCTs of 4,000 K or 6,500 K. To date, 15 healthy subjects have entered into two separate studies and completed over 150 nighttime melatonin suppression studies. The majority of plasma samples collected in these studies have been assayed for melatonin content.

2. Evaluate selected polychromatic LED stimuli for supporting visual performance, color discernment, and neuroendocrine potency inside replicas of ISS Crew Sleeping Quarters (CSQ). The replica CSQs are illuminated by the ISS prototype solid state lamps (SSLM-R). To date, 16 healthy subjects have participated in two separate nighttime melatonin suppression studies within replicas of the CSQs illuminated by white SSLM-R light with a CCT of 4,800 K. The majority of plasma samples collected in these studies have been assayed for melatonin content. In addition, two studies were done on visual performance and color discernment within replicas of the CSQs illuminated by different polychromatic light stimuli emitted by SSLM-Rs. Analysis of data from these studies is in process.

3. Compare the potency of ambient light on the surface of Earth and Mars for melatonin regulation. In collaboration with Peter Smith, Ph.D., of the Lunar and Planetary Laboratory at University of Arizona (UA), optical elements were developed and installed for simulating Earth daylight in a Ganzfeld dome. To date, 9 astronaut-aged, healthy subjects have and completed 78 nighttime melatonin suppression studies relative to Ganzfeld simulation of Earth daylight. In parallel, the engineers at UA have developed a newer exposure system that simulates ambient Earth, Moon, and Mars light in a larger, non-Ganzfeld environment. Currently, this equipment is being installed in our laboratory for radiometric and photometric assessment.

Together, the research from the two prior funding cycles along with the studies performed in the past year provide significant progress towards the development of lighting countermeasures for sleep and circadian disruption in astronauts and ground crew members. Programmatically, it has provided information that was used in the revision of Constellation Program Human-Systems Integration Requirements (December, 2009). Further, the results will also impact the NASA Human Integration Design Handbook and the Space Flight Human Systems Standard, NASA-STD-3001 that provide guidance for crew health, habitability, environment, and human factors in human spaceflight. Our progress addresses Critical Risk areas 9 (EVA 7) and 22 (Sleep 5, 9, and 10) in the NASA Human Research Program Integrated Risk Plan (2009). These areas concern countermeasures that will optimally mitigate performance problems associated with sleep loss and circadian disturbances and the "mismatch between crew physical capabilities and task demands."

Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The knowledge we hope to gain from this research, though focused on spaceflight, will also benefit people on Earth. The sleep deficits experienced by astronauts during spaceflight can be considered a threat to the success of space missions (Bioastronautics Roadmap, Longnecker and Molins, 2005; NASA Human Research Program Integrated Risk Plan, 2009). The resulting physiological and behavioral changes caused by sleep and circadian disruption can lead to diminished alertness, cognitive ability and psychomotor performance (Dijk et al., 2001). As a measure to counteract sleep disruptions, over 45% of all medications taken in space are sleep aids (Putcha et al., 1999). A significant portion of the global population suffers from chronic sleep loss and/or circadian-related disorders. Evidence for disease or illness occurring due to a disruption of circadian homeostasis has mounted significantly in the past several years. In the United States, nearly 22 million Americans do shift work that interferes with a biologically healthy nocturnal sleep cycle (US Bureau of Labor Statistics, 2007). Shift workers have been shown to be more likely to suffer from a wide variety of ailments, including cardiovascular disease, gastrointestinal distress, and cognitive and emotional problems. Furthermore, epidemiological studies of female shift workers have shown that they are statistically more likely to suffer from breast and colon cancer as compared to their day shift counterparts. The World Health Organization has identified shift work as a probable risk for cancer (The International Agency for Research on Cancer, 2007). Our laboratory is involved in testing the hypothesis that nighttime exposure to light suppresses melatonin and contributes to cancer risk (Blask et al., 2005; Stevens et al., 2007). Aside from evidence of a breakdown in physical health, the effects of circadian disruption and sleep loss are known to have potentially dangerous behavioral effects. Mental fatigue, diminished alertness, loss of psychomotor coordination and decreased physical performance are all commonly found in individuals with sleep loss, sleep debt, or circadian misalignment. The impact of these dangers affects many industries, including transportation, manufacturing, communications, and medicine. It has long been a source of concern for the military and, more recently, homeland security. Many people also experience the same effects after air travel across several time zones. In the past, the U.S. Air Force has supported our laboratory to study the acute alerting effects of light (French et al., 1990; Brainard et al., 1996). Our current work for NIH has continued this effort (Lockley et al., 2006). Existing light therapy interventions stand to benefit from enhancing our understanding of how different wavelengths of the spectrum affect human circadian and neurobehavioral regulation. The result may be a more efficient intervention with increased potency and/or fewer side effects. One such disorder currently being treated with bright white light is Seasonal Affective Disorder (SAD), also known as winter depression. It is estimated that as many as 1 in 5 Americans suffer from SAD or its milder version, subsyndromal Seasonal Affective Disorder (sSAD) (Lam and Levitt, 1999). Similar bright white light interventions are also used to treat jetlag. Side effects from exposure to bright white light for these and other therapies include: hypomania, headache, vision problems, nausea, dizziness, and anxiety. Optimizing the light spectrum for specific affective and/or circadian-related disorders may deliver the same medical benefit with lower levels of light intensity, and potentially fewer side effects. Our group has completed Phase I testing of a lower intensity of light therapy with blue solid-state lighting for SAD patients (Glickman et al., 2006).
Task Progress:	The goal of our work is to optimize lighting that supports vision and serves as a circadian countermeasure for astronauts and ground crew during space missions. One aim is to determine the neuroendocrine potency of light emitted by the white LEDs being considered by NASA for in-flight and habitat lighting systems. Previously, we initiated a melatonin suppression study with young, healthy subjects (N> or =7) using two 4' by 4' white LED lighting systems (CCT of 6,500 K). All 83 experimental nights and their melatonin assays are now completed along with the pupillary dilation statistics. Preliminary calculations of retinal irradiances and melatonin suppression curve fits have been done. Recently, two different 4' by 4' white LED lighting systems (CCT of 4,000 K) have been designed and installed in our lab by The Lighting Sciences Group. Young, healthy subjects (N> or =8) have been entered into a melatonin suppression study with this new equipment and have completed over 70 experimental nights. A majority of the plasma samples collected have been assayed but no statistical analyses have been done for this ongoing study. As part of a separate project (NASA #NNX09AM68G), we developed both high-fidelity and low-fidelity replicas of ISS Crew Sleeping Quarters (CSQ). We are testing acute visual and neuroendocrine responses within these replicas illuminated by the ISS prototype lamps (SSLM-R). In one study, 8 healthy, young subjects completed 40 experimental nights for a melatonin suppression study in the high-fidelity CSQ. Melatonin assays and statistics are complete for this study. Statistics on pupil dilation and retinal irradiances have not yet been completed. Four of these same subjects participated in a study on visual performance and color discrimination in the same CSQ lighting conditions. A separate study on visual performance and color discrimination was conducted with astronaut-aged subjects (N> or =8, 39.0 +/- 3.1 years) inside the high fidelity CSQ with three polychromatic light stimuli emitted by the SSLM-R as well as two control conditions. All test runs have been completed and data analysis is in process. Working with a low-fidelity CSQ, 8 healthy subjects (25.6 +/- 1.2 years) have been entered into a melatonin suppression study with a full range of intensities of white light (CCT 4800 K) emitted by the SSLM-R. Over 70 exposure nights have been completed and a majority of the study blood samples have been assayed. One of our aims is to compare the potency of ambient light on the surface of Earth, Moon, and Mars for acute melatonin suppression, since circadian and sleep regulation depends on an astronaut's exposure to both artificial light and passive ambient illumination. Working with the Lunar and Planetary Laboratory, a new lighting system has been completed to simulate extraterrestrial light. This equipment is currently being installed in our laboratory and should be available for radiometric assessment and human studies in the coming year.
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